Rimsha Afzal

AI Product Analyst — Junior Machine Learning Engineer

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About Me

Master's student in Artificial Intelligence Systems at the University of Trento with a background combining technical AI experimentation, NLP research, and user-centered design. I enjoy working at the intersection of technology and problem understanding, where technical solutions need to be shaped around real user needs. My strengths lie in combining hands-on technical work with strong communication and stakeholder engagement, allowing me to translate complex AI capabilities into clear and practical solution ideas.

Education

University of Trento, Italy 2025 – Ongoing

Master's Degree in Artificial Intelligence Systems

Relevant coursework: Machine Learning, Natural Language Processing, Deep Learning, Autonomous Software Agents, Designing Large Scale AI Systems

University of Trento, Italy 2021 – 2025

Bachelor's Degree in Computer Science

Relevant coursework: Algorithms and Data Structures, Software Engineering, Human-Computer Interaction

Projects

Transformer-Based NLP Pipeline for Complaint Urgency Detection Bachelor's Thesis

- Designed and evaluated an **NLP pipeline** for prioritizing financial complaints by combining **complaint classification** and **sentiment analysis**
- Preprocessed and split a real-world complaint dataset, then tokenized complaint texts using **BERT-based representations**
- Fine-tuned **BERT** and **FinBERT** models for complaint-type classification and managed systematic experimentation on an **HPC cluster using Slurm**
- Compared multiple pretrained transformer models for sentiment analysis, including **FinBERT-tone, Twitter RoBERTa, multilingual BERT, and DistilRoBERTa**
- Combined classification outputs and sentiment signals into an **urgency metric** to explore a practical decision-support pipeline for complaint prioritization

OpenMove Internship — Customer-Care Workflow Design

- Investigated workflow pain points and operational needs through **stakeholder interviews** in the context of a customer-care handling solution
- Translated qualitative input into **design requirements**, identifying how needs should be reflected in navigation flows, screen logic, and functional choices
- Designed **Figma mockups** and an **interactive prototype** to turn early requirements into a tangible solution concept
- Defined interface choices with attention to **usability**, **clarity of actions**, and support for the underlying customer-care process

Start-up Lab — Idea Validation and MVP Design

- Led a team project through the early stages of start-up development, from **idea generation** and **problem validation** to **MVP definition**
- Conducted **external interviews** to test assumptions, gather feedback, and assess whether the identified problem was relevant and worth solving

- Contributed to translating early ideas into a more concrete solution concept supported by an **MVP mockup**
- Prepared and delivered presentations communicating the project's logic, development path, and business potential
- Supported early feasibility assessment through **basic cash-flow reasoning** and sustainability considerations

Community Engagement

Italian Tech Week Volunteer 2025
Turin, Italy

- Supported event operations during a 3-day international technology conference
- Managed front-desk guest reception and participant registration
- Observed event organization and logistics behind large-scale tech conferences
- Attended talks from industry leaders on entrepreneurship, technology, and innovation

Current AI Projects

Multimodal Image Retrieval with CLIP Architectures
Exploring multimodal image retrieval through vision-language embeddings and similarity conditioning in CLIP-based architectures.

Planning-Based Autonomous Software Agents
Studying autonomous agent design through planning, belief management, and decision-making in dynamic environments.

Optimizing Transformer Language Models
Exploring GPT-2 optimization through hyperparameter tuning, architectural changes, and parameter-efficient fine-tuning with LoRA.

Technical Skills

AI & Machine Learning
Machine Learning, Natural Language Processing, Deep Learning, Transformer Architectures, Language Modeling

Programming
Python, Java, HTML, CSS, Kotlin

ML Frameworks & Libraries
PyTorch, HuggingFace, Transformers

Tools
Git, Figma, HPC (Slurm)

Product & Research Skills
Problem Framing, Requirement Translation, Stakeholder Interviews, User Research, Prototyping

Languages

English — C1
Italian — C1
Urdu — B2

Portfolio